

REPORT: TASK 2, 3, and 4

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Task 2: Analyzing fairness and bias in the Census Income Dataset

In this task, I take a look at fairness and bias in income prediction model using Adult Census Income dataset. In particular, it analyzes machine learning techniques aimed at the task of classifying individuals in terms of income groups ($\leq 50K$ and $> 50K$ annually), comparing their performance, strategies for bias mitigation, effects on equity, and effects on accuracy.

Dataset Overview

The Adult Census Income dataset predicts whether the person earns $\leq \$50K/\text{year}$ or $\$50K+/\text{year}$ based on certain characteristics.

[7]:

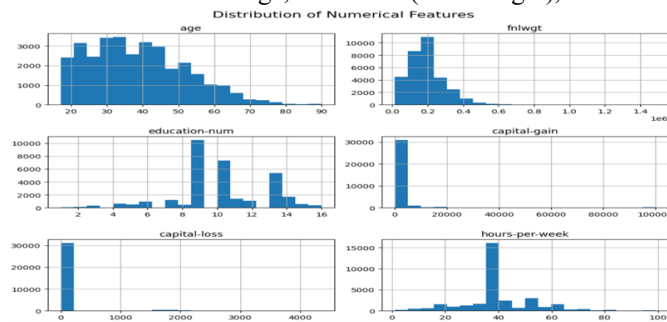
	age	fnlwgt	education-num	capital-gain	capital-loss	hours-per-week
count	48842.000000	4.884200e+04	48842.000000	48842.000000	48842.000000	48842.000000
mean	38.643585	1.896641e+05	10.078089	1079.067626	87.502314	40.422382
std	13.710510	1.056040e+05	2.570973	7452.019058	403.004552	12.391444
min	17.000000	1.228500e+04	1.000000	0.000000	0.000000	1.000000
25%	28.000000	1.175505e+05	9.000000	0.000000	0.000000	40.000000
50%	37.000000	1.781445e+05	10.000000	0.000000	0.000000	40.000000
75%	48.000000	2.376420e+05	12.000000	0.000000	0.000000	45.000000
max	90.000000	1.490400e+06	16.000000	99999.000000	4356.000000	99.000000

They were ranging in age from 17 to 90 years, in moderate variability of education level pegged to the average of 10.08, and mainly in the education level from 9 to 12 with variability of weights (FNLWGT). Because most of the values are 0, suggesting infrequent monetary fluctuations, the frequency distribution of capital gain or loss is highly right skewed. The average employee lives 40–45 hours a week, which is easy to explain by working time regularity. The financial features might be dependent and biased: this might need special treatment while examining.

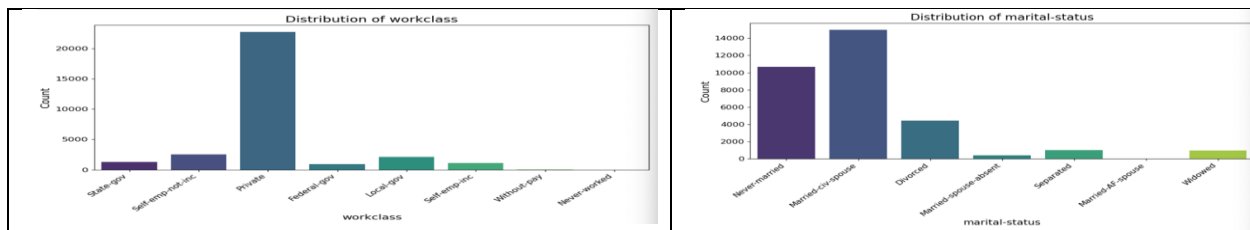
Numerical Features and Categorical Features

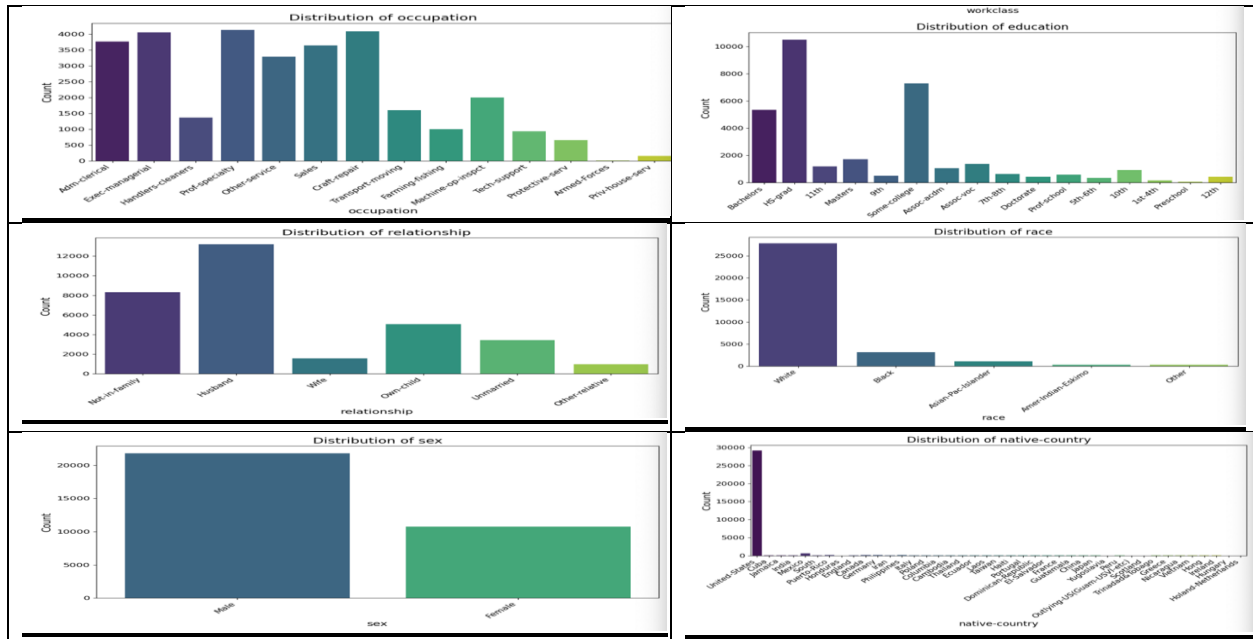
The numerical features are six, and the categorical features are nine.

Numerical features are age, FNLWGT (final weight), education number, capital gain, capital loss, and hours per week.

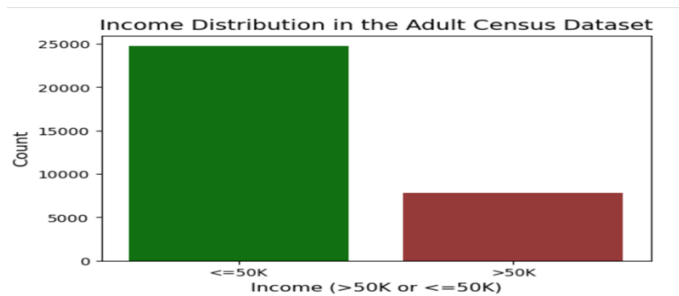


The categorical features are class, education, marital status, occupation, relationship, race, sex, native country, and income (target variable).





Analyzes and visualizes the distribution of the target variable, income



There is a clear imbalance in the dataset, with approximately 3 times more individuals in the <=50K income category than in the >50K category.

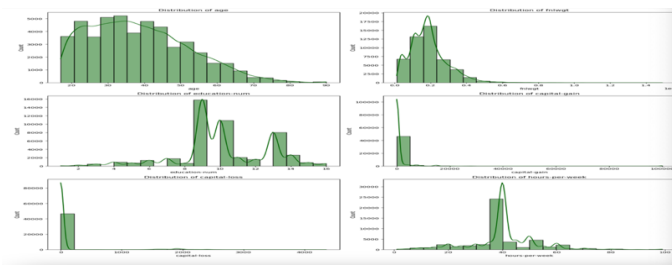
Missing Values

The yellow lines show missing values, mainly in the workplace, occupation, and native country. If not addressed, these could affect analysis accuracy.



Data skewness and bias Skewness in Numerical Features

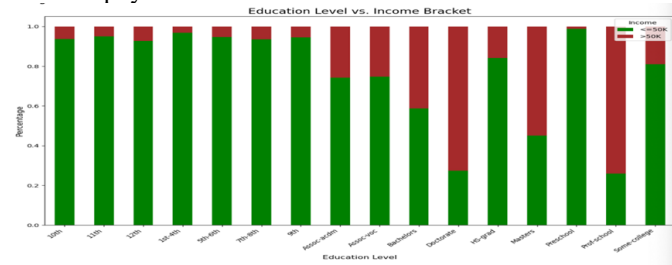
Skewness in numerical features shows distribution variations: Age (0.56), float (1.44), and hours-per-week (0.24) are positively skewed, while education-num (-0.32) is slightly negatively skewed. Capital gain (11.89) and capital loss (4.57) are highly right-skewed, indicating significant outliers.



The distributions show skewness, especially in capital gain and loss, which may impact model performance if unaddressed.

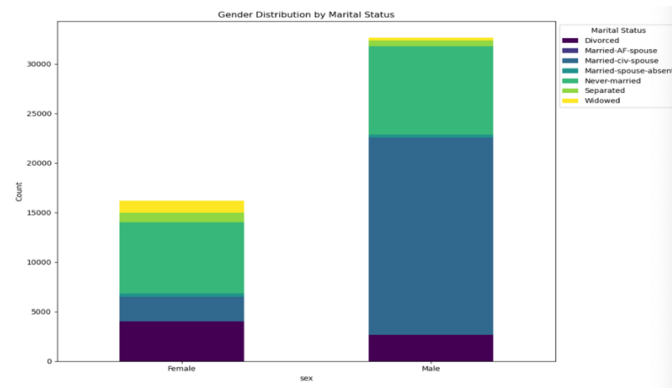
Evaluating the relationship between education level and income

Examine physical health and mental health across four educational levels.



Gender Distribution across Different Marital Status

This underlines gender distribution about marital statuses and demonstrates such frequent categories as “male and married” and “female and not married.”

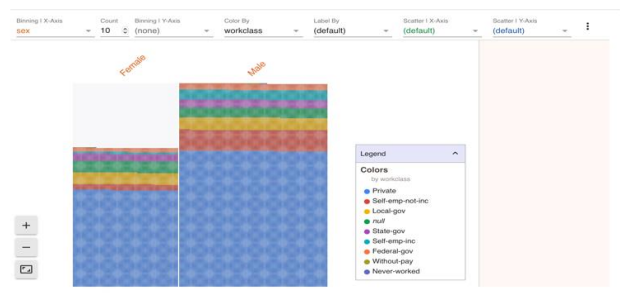


The gender by marital status of the population reveals that most males are married, while the female population is almost evenly divided between the married, the never-married, and the divorced. This implies that gender does indeed influence marital status.

Additional Tool: Facets Dive and Facets Overview

For further analysis of the collected dataset, the Facets Dive tool was used to examine unique values of the specific data point with the ability to alter the parameters of visualization. It offers much better insight into data distributions and outliers that may harm the sample bias.

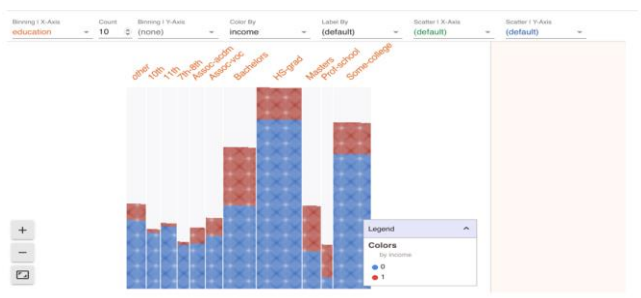
Bivariate analysis



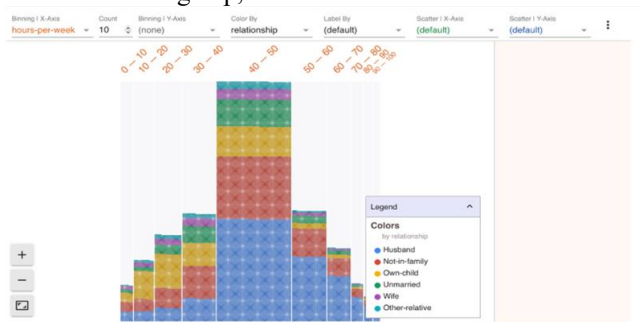
Of all the work classes, The Private work class is most prominent in both male and female employees, especially in the male employees where there is more variation in the types of work they are engaged in, such as self-incorporated businesses and local government.



As for the income, the White race is predominant both in the income category >50K and in the category <50K income. It is also clear that Blacks and Amer-Indian-Eskimo are largely in the <=50K income range, suggesting an income split between different races.

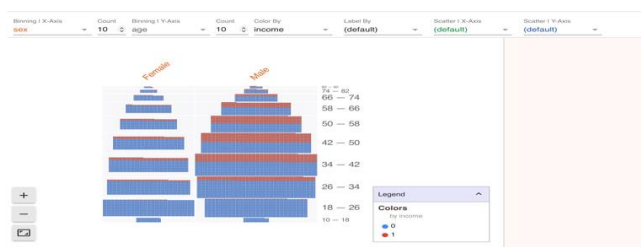


Education level like bachelors, master, prof-school contains higher population in > 50000 income group then lower education level 10th 11th HS-grad contains lower population in <=50000 income group. This goes to show that as the level of education go up, so also does the income level.

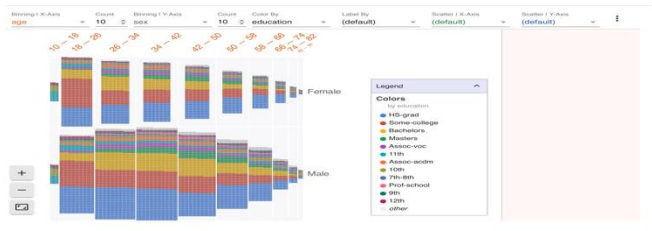


More workers who work for 40 hours a week are Husbands (blue) than Not-in-family and unmarried groups. Employees working fewer or more hours are rare and can be found in every relationship stage.

Multivariate Analysis

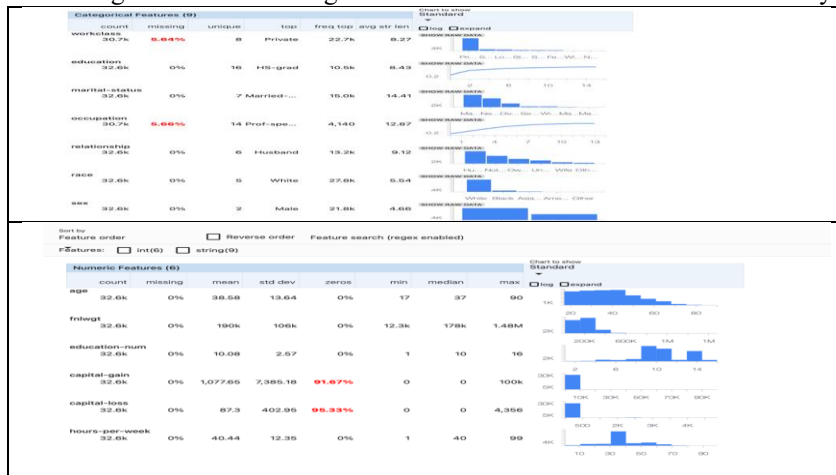


Males have a higher percentage in the >50000 income group within different age groups, while females dominate the <= 50000 income group. They are categorized by income; there are visible layers of current income inequality, especially for males 34-58 years old, where females are much lower in the higher income in comparison with males.



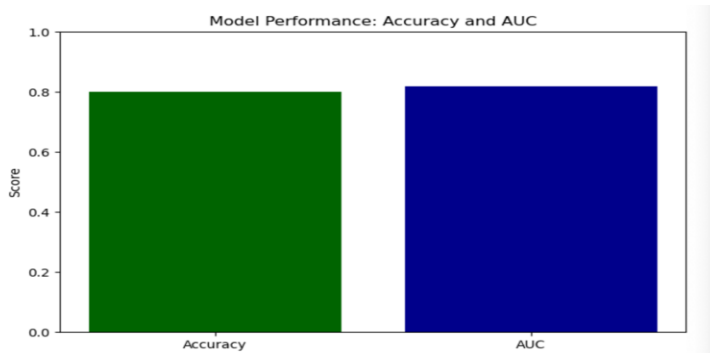
The result also shows that a significantly higher proportion of males have attained higher education levels of Bachelors and Masters than females, more especially in the 26–50 age group. Males and females have more dominance in mid-tier education levels such as HS-grad, In Some colleges, younger and older persons have less diverse education levels.

The Facets Overview was also used to create basic statistical computations about the dataset and its various features that are significant in detecting distributions and relations in an additional layer on facets for fairness.



Exploring Fairness When Training Models

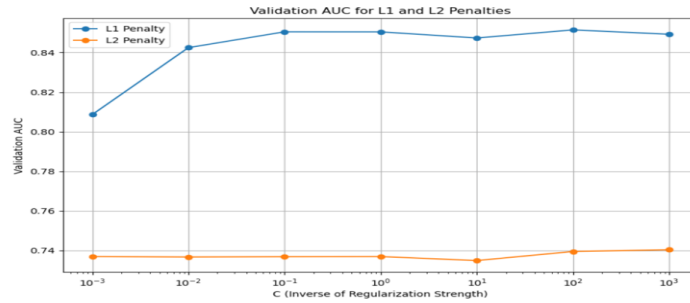
This task concerns fairness issues in the Adult Census Income model, and deals only with the “sex” feature. First, the data were preprocessed to enforce clean data set, and categorical data were recoded to numerical data set. The other evaluation measures, including Mean Difference (-0.199), and Disparate Impact (0.363), also confirmed this bias in the model against females. A Logistic Regression model was built of the data and a high accuracy was obtained but this depicted the need to correct the bias to give better results on all samples.



Hyperparameter Tuning for Logistic Regression

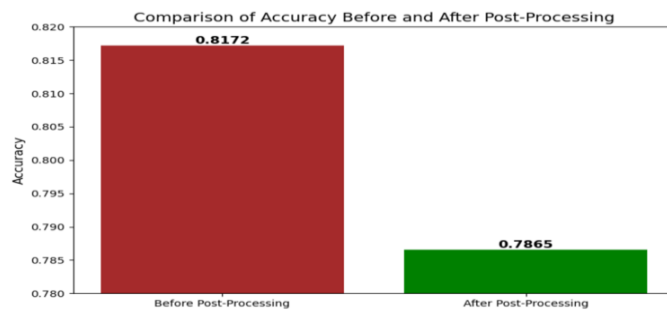
The final hyperparameter tuning revealed that `LogisticRegression(C=100, max_iter=1000, penalty='l1', solver='liblinear')` is the final model with fairly good test AUC 0.857 in our model. The L1 penalty was superior to L2 in all folds, achieved a better feature selection by offering the highest validation AUC rates. From the result, the optimal value of C is 100 and both accuracy and regularization were adequate at this point. On the other hand, the L2

penalty produced inferior results at every C value the experiment was conducted on. These results demonstrate subject-to-subject penalty selection and tuning for improving model performance and feature relevance.



Evaluating Bias in Predictions using different approaches

The model also presents a bias toward females even though all the variables have passed the statistical significance test. Since all the variables have a $p < 0.05$, the mean difference of -0.176 and disparate impact of 0.173 are far below the fairness threshold. Statistical analysis revealed significant gender differences were identified where females had a false negative rate ratio of 1.56 and males false positive rate ratio of 0.17. Post-processing reduces discrimination much higher as compared to pre-processing and has better F-measure and EPPS than accuracy rates, making it the most efficient method of fairness.



Metric	Without 'sex'	Reweight Data	Post-Processed Data
Validation AUC	0.8461	0.8463	
Test AUC	0.8522	0.8523	0.9533
Training AUC		0.8432	0.9533
Accuracy	0.82	0.816976	1.0
Mean Difference	-0.140130	-0.140105	-0.043200
Disparate Impact	0.286602	0.288702	0.598040
Error Rate Difference	-0.102963	-0.102076	-0.140540
False Negative Rate (Privileged)	0.532163	0.530424	0.744263
False Negative Rate (Unprivileged)	0.750748	0.749751	0.743769
False Negative Rate Ratio	1.410748	1.413493	0.999336
False Positive Rate (Privileged)	0.072584	0.072584	0.039822
False Positive Rate (Unprivileged)	0.031395	0.031909	0.039501
False Positive Rate Ratio	0.432527	0.439618	0.991926

Conclusion

From this analysis, biases in the Adult Census Income dataset, such as low diversity about gender and race and association between income, gender, and education levels as presented by the dataset, fall under stereotyping biases. Facets visualization identified further feature distributions that may cause unfair outcomes in employment or credit rating, for example. RP-reweight and equalize achieved a decent overall de-biasing without significant performance degradation, and post-process was the best balance between fairness and accuracy. Problems with indirect correlations are the reason that removing sensitive attributes such as sex is insufficient.

Task 3: Interpretability of ML Model

The number of samples in this data set, which is used a lot in the machine learning community, is called the Heart Disease dataset of UCI, which provides every possible aspect of health to predict the possibility of heart diseases. This task is involved in the evaluation of the interpretability of a logistic regression model developed from the Heart Disease UCI data set. The activity comprises model construction, model evaluation, SHAP values analysis, and feature importance analysis. We also plan to describe how each feature influences model predictions and what that means for using such a model in healthcare.

Dataset Overview

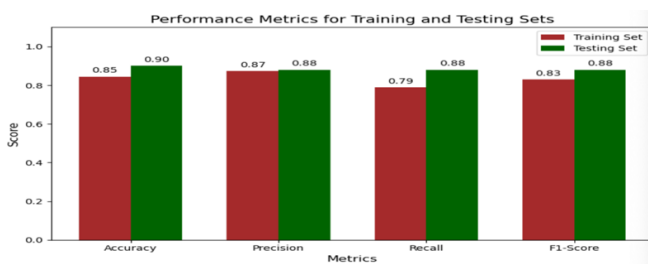
The Heart Disease UCI dataset contains 76 instances and 14 features, including traditional. The Heart Disease UCI dataset has 76 records and 14 domains of features with clinical data and exercise profiles as the important measure. It aims at determining whether or not a person has heart disease.

Data Preparation and Model Evaluation

We loaded the Heart Disease data, where features for age and cholesterol and other features were preprocessed, where we dropped missing values and normalized them. The data was divided into 70:30 & 80:20 for training and testing, respectively. For classification, ward survival was considered to be a dependent variable: 1 for heart disease, 0 for no heart disease, and logistic regression was used. Here, the descriptors of accuracy, precision, recall, and F1-score were used, and the values were seen to be satisfactory in predicting the new heart disease accurately.

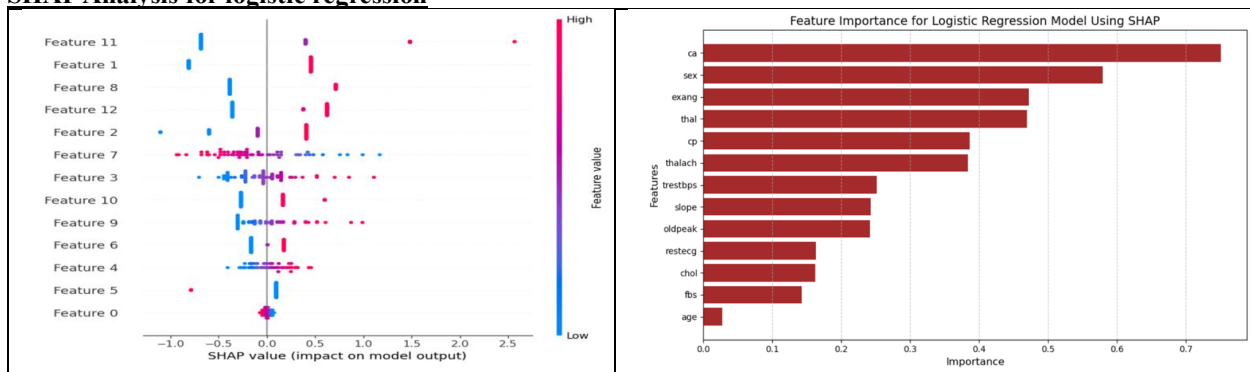
Interpretability Analysis of Logistic Regression on Heart Disease Prediction

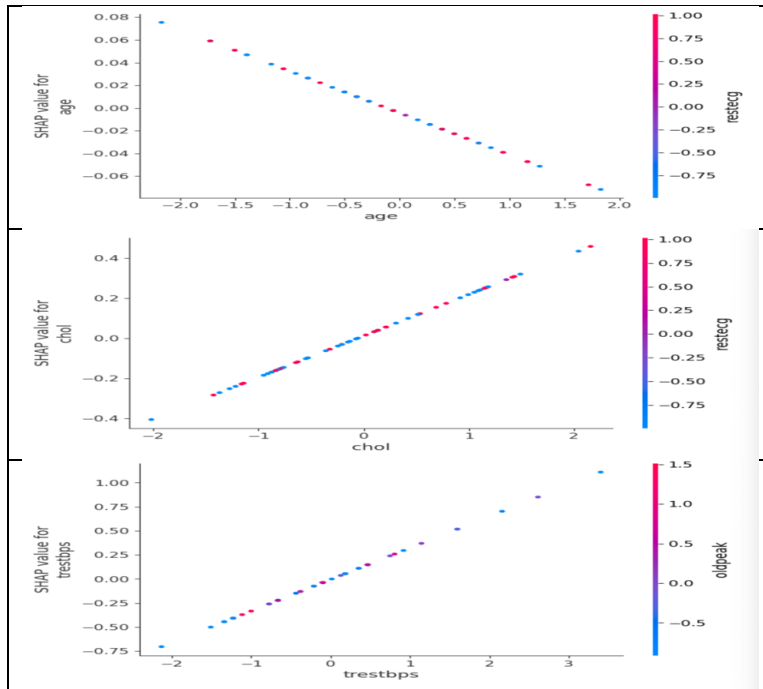
Logistic regression is applied to predict heart disease from the UCI database. To evaluate model interpretability, SHAP explanations, and feature importance are applied to see how the underlying model makes its decision.



Good generalization is observed for the model where the metrics of training and testing are similar. Error analysis is slightly better than training accuracy with values of 0.90 and 0.85 respectively. All the forecasts are fairly and symmetric, high sensitivity, specificity, recall rates, and F1-scores. The reproducibility of these results shows that there is no issue of overfitting or underfitting, the stability of the model.

SHAP Analysis for logistic regression

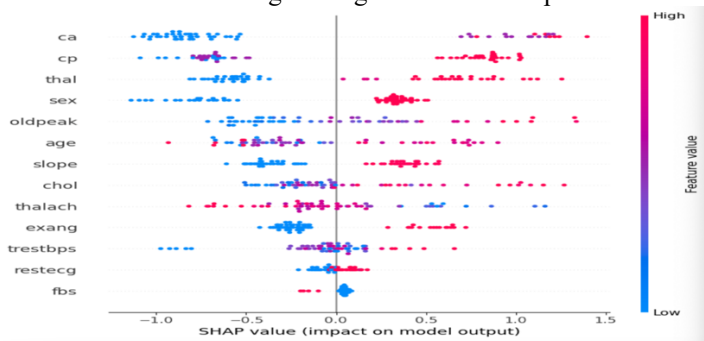




The results of the SHAP analysis reveal that the most important features affecting the heart disease risk are age, cholesterol, and blood pressure. It is clear that adverse factors such as older age, high cholesterol, and high blood pressure increase risk, and favorable factors that are the opposite of these decrease it. It is found to be dependent on first-order interactions concerning age and cholesterol and second-order for old peak for blood pressure, of which high combined trestbps and oldpeak are indicative of risk. SHAP makes the data safe by encapsulating the identity of its users, manages to eliminate misuse, and provides better interaction analysis than logistic regression, which provides only simple linear relations.

SHAP Analysis for XGBoost

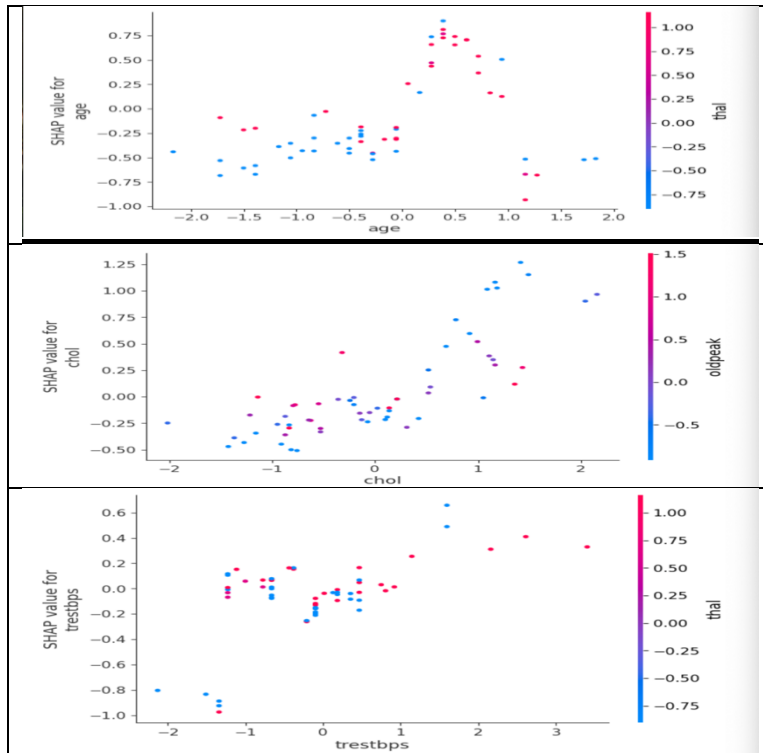
It is clearly shown that the proposed XGBoost model to improve the performance obtained the 86.7% accuracy and 0.924 ROC AUC score. SHAP values elucidated the feature importance to explain the Age, Cholesterol, and Blood Pressure to be the key predictors. :Positive relationship outcomes were identified for high cholesterol and high blood pressure. It was better to logistic regression and also provided the anonymity of the data.



The level of risk for heart disease increases with age, and the majority of the people afflicted by it belong to the older generation. The effect of cholesterol depends, while high systolic pressure enhances the threat of heart disease.

SHAP Dependence Plots

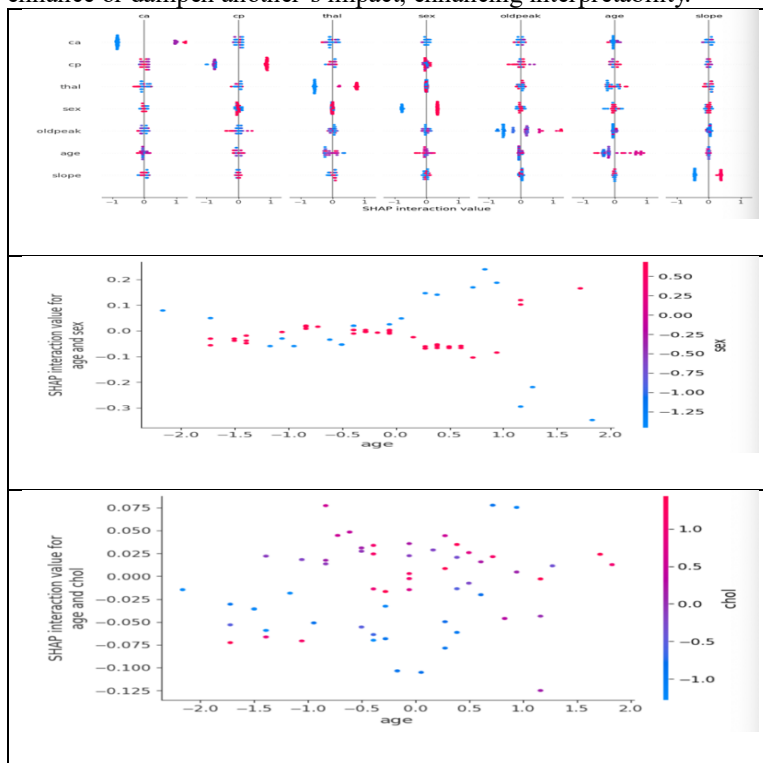
In SHAP, coefficient dependence plots are depicted, illustrating the effect of features and their interactions on the model output for high and low values.



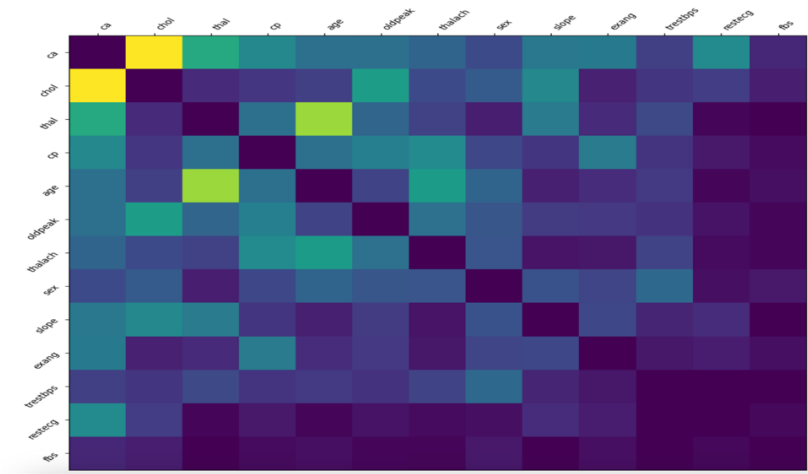
SHAP plot shows that features such as Age (50-60), Cholesterol (200-300), and Trestbps (high systolic pressure) are the most influential in the heart disease predictions and feature interactions like cholesterol & “oldpeak.”

SHAP interaction values

The SHAP interaction values, therefore, define how feature pairs contribute to the prediction, which could either enhance or dampen another’s impact, enhancing interpretability.



Values of the interaction with the SHAP emphasize the most important pairs of features, such as “ca,” “chol,” and “thalach,” which affect the result. Again, while age interacts more linearly with sex for males and non-linearly for females, age and cholesterol have the largest overall interaction for ages 50-59.



The coefficient analysis has revealed a negative relationship between thalach and age, as for chol and trestbps, the link is positive. From Table 5, newly developed characters, including Age and old peak, are getting more important with increasing age and blood pressure, while sex, slope, and FBS reveal a low correlation, suggesting independence.

Results of Models

Criteria	Logistic Regression	XGBoost
Performance Metrics	Training Accuracy: Accuracy: 0.85 Testing Accuracy: Accuracy: 0.90	Evaluated with Harrell's C-Index. Accurate and reliable ranking of predictions.
Key Features	Ca, sex, Fbs were most important. Lesser impact from thalach, age, and chol.	Age, chol, trestbps were the most influential features.
SHAP Analysis	Identified linear feature impacts. Cholesterol peaked at 200-300; age significant >50.	Captured nonlinear trends. Highlighted critical impact zones for age and cholesterol interactions.
Feature Interactions	Not analyzed.	Strong interactions like age and cholesterol and cholesterol and age and sex showed key combined effects.
Complexity	Simple and interpretable.	More complex but better at capturing nonlinear relationships and interactions.

Conclusion:

The interpretability analysis explained the result of the model prediction, the features consisting of age, cholesterol level, and systolic blood pressure that are significant in the prediction of heart disease. As seen from the case provided in the study, the analysis using SHAP was indeed beneficial as it not only allowed one to identify what it is that the model is doing but also pointed out the various patterns that were required for the identification of the solution space. This ensures that the interpretability of the results greatly enhances the reliability of the model and its ability to afford quality results that can explain the impact of the features apart from improving its ability to deliver quality results, especially when deployed in very sensitive sectors such as healthcare.

Task 4: Privacy

Privacy is critically important for such websites as Amazon.co.uk, LinkedIn.com, Google.com, and Facebook.com, which have their users' information used to provide tailored services as well as marketing materials. It also revisits the data of the two regarding privacy, the significance of privacy in machine learning, and the way that techniques such as differential privacy can protect the data of a user.

AMAZON

1. Privacy and the need for privacy-preserving machine learning

Data Collected by Amazon.co.uk:

1. Purchase History:
 - Registers things purchased with actual cash, merchandise, or in the form of downloads (i.e., books, films, or magazine subscriptions).
 - For recommending the predefined order type as well as for returns or order type history in case of storing the data. [1]
2. Browsing Habits:
 - Records specific web entrance, items seen, and classes of viewed items of a customer.
 - Allows Amazon to show preferred interest advertisements and recommend products to the customer. [1]
3. Payment Information:
 - Contains credit card/debit card details, billing addresses, and preferred payment methods.
 - Guarantees safe payment and more convenient and quicker future shopping. [1]
4. Shipping Details:
 - Store delivery addresses and the contact number for order arrangements.
 - Offers information about the order and allows you to select quicker shipping methods. [1]
5. Search Queries:
 - Supervises characters typed into the search bar to filter results regarding product search.
 - Enables the determination of trends that are most popular for ads to be posted to relevant users. [1]
6. Device Information:
 - Gathers data on devices being used (for example, brand, operating system, net addressing).
 - Improve the security of use and make the usage tailored to the specific devices. [1]
7. Alexa Interactions:
 - Stores voice instructions and interaction with Alexa gadgets.
 - Enhances the AI and offers a product from the user's history (lists or questions asked). [1]

Importance of Privacy:

Machine learning is applied at Amazon to provide better recommendations to customers, to efficiently deliver products, and to provide targeted ads. This is important so that one does not fall into the hands of hackers who may be out to steal the user's identity or rip off their credit card information or anything that the user inputs through the application.

Potential Impact of Privacy Breaches:

- Individual Users: Sharing personal purchase records or recordings can expose individuals to identity theft or financial scams.
- Society: Massive data losses may cause a loss of consumer confidence in digital purchasing services and reduce shopping on the Internet.

Case Study:

Research conducted at UC Davis showed that Amazon's Alexa collects user data and uses that data for advertising purposes. The practice elicited a lot of concern for the privacy of users. [1]

2. Achieving the Four Pillars of Privacy Preservation

Four Pillars of Privacy-Preserving AI in Amazon

1. Training Data Privacy: Amazon chooses moderate capacity usage and additionally conceals personal information while training its natural language understanding models.
2. Input Privacy: In terms of data security, there are good methods of data input encryption and anonymization applied.
3. Output Privacy: They apply differential privacy to make sure that the results of their models cannot violate the privacy of an individual piece of data.
4. Model Privacy: To improve the privacy of the models, more work has been conducted and is being conducted on privacy-preserving machine learning, for instance, federated learning. [2]

Recommendations for Improvement:

1. Federated Learning: Introduce federated learning as a way of training models across devices without having to aggregate data and risk its privacy.
2. Homomorphic Encryption: Use exact execution to perform computations on encrypted data, leaving data entirely safe even in the course of computation.
3. Secure Multi-Party Computation: Develop a method of secure multi-party computation that would enable several parties to carry out computations on data that was shared between the parties, but the actual data was not disclosed to other parties. [3]

3. Differential Privacy in Machine Learning

Differential privacy has been considered by Amazon as the solution for improving the privacy of its machine-learning models. For example, they have studied the use of differential privacy, which is a fundamental concept within this neural network; for instance, adding noise to the training examples to ensure they become more accurate.

Benefits:

1. Enhanced Privacy:
 - Protecting User Data: Differential privacy essentially involves adding random noising to the data so that it becomes quite impossible to reel back from the data your purchase history and your voice commands, among other identifiable information.
 - Preventing Misuse: The chance of hackers or any other person predicting individual user information from such a modeling machine is highly eliminated. [4]
2. Regulatory Compliance:
 - Following Privacy Laws: That way, it can allow them to stick to provisions such as GDPR or CCPA while aggregating data that doesn't identify the users in question.
 - Easy to prove Compliance: The noise-added process is fully auditable, and it can be shared with regulators to ensure compliance with privacy requirements. [4]

Trade-offs:

1. Model Accuracy:
 - Reduced Precision: Incorporation of noise into data or model outputs results in a slightly lower accuracy in predicting, for instance, recommendations for products or ads to be posted.
 - Impact on User Experience: Less accurate means there remains a possibility of giving the customer less relevant recommendations—hence reduced satisfaction.
2. Complexity:
 - Balancing Privacy and Utility: Deciding the correct amount of noise required for the model proves to be a tricky task. This is because excessive use of noise limits the model's usefulness, while the imposition of little noise will undermine user privacy.
 - Technical Challenges: It should be mentioned that differential privacy, which is a significant advantage in the present problem, requires resources and especially expertise to run the trade-off properly.

Consequently, more efforts have been made in Amazon to ensure that privacy-preserving approaches are implemented within the process of machine learning and to generate greater attention and improve innovative techniques to match users' satisfactory privacy expectations and privacy regulation standards.

LINKEDIN

User data plays an important role in sites such as LinkedIn.com by providing large amounts of user data for improving services via ML.

1. Privacy and the need for privacy-preserving machine learning

Data Collected by LinkedIn.com:

1. User Profiles:
 - It can contain name, address, skills or experiences, or any other information regarding a particular candidate.
 - Assists LinkedIn to notify you of opportunities and ensure it matches you to the right opportunities.
2. Connections and Activity:
 - Monitors one's professional network, likes, comments, and shares, as well as entails the option of following posts or groups.
 - Precisely invite them to a connection to make them aware of the activity that is worth their attention in your network.
3. Resumes:

- Businesses with uploaded resumes for job employment opportunities or other purposes apart from job-seeking.
- It was to be employed in pairing users and job posts and sharing such with the recruiters.
- 4. Search Queries:
 - Records the search terms that are used in a search of jobs, firms, or other persons.
 - It helps LinkedIn to display better search results, enable suggestions, and also recommend better jobs.
- 5. Messages:
 - Stores copying of messages you send and receive on the platform directly.
 - Provides safety in the communication procedure, and it can employ data to recommend proper replies or even ideas.

Importance of Privacy:

LinkedIn utilizes ML algorithms to provide personalized job recommendations, network suggestions, and targeted advertising. Protecting user data is essential to prevent unauthorized access, misuse, or breaches that could lead to identity theft, professional harm, or loss of user trust.

Potential Impact of Privacy Breaches:

1. Individual Users: Publishing personal information, a contact base, or messages can result in identity theft or problems in one's career.
2. Society: The loss of databases containing people's data decreases the population's confidence in the professional utilization of Internet platforms.

Case Study:

Finally, in September 2024, the media informed its users that LinkedIn was refuting data to train AI models without informing the people concerned. [5]

2. Achieving the Four Pillars of Privacy Preservation

Four Pillars of Privacy-Preserving AI on LinkedIn

1. Training Data Privacy: Controlling methods that are currently applied in LinkedIn include PriPeARL for privacy-preserving analytics and reporting, adapted from differential privacy. [6]
2. Input Privacy: Clean data is also protected using encryption and anonymization of the inputs entered by the user.
3. Output Privacy: LinkedIn uses the principles of privacy-preserving data analytics systems as a means to guarantee that the results that it delivers do not cause insecurities in individual data. [7]
4. Model Privacy: Current work seeks to explore more methods to promote privacy-preserving machine learning models. [8]

Recommendations for Improvement:

1. Federated Learning: Implemented every architecture with federated learning to enable the training of models across various devices while not compacting data in one location, thus increasing the data security level. [9]
2. Homomorphic Encryption: Homomorphic encryption should be built and applied in a way that shows encrypted data is safe from unauthorized access even when the computations are being made to the collected data. [9]
3. Secure Multi-Party Computation: To enable several parties to compute functions on their data with the data of the other parties but without revealing it. [9]

3. Differential Privacy in Machine Learning

LinkedIn has been looking into the use of differential privacy to increase the privacy of their machine learning model. For instance, they have proposed to study a method to make the privacy-preserving neural networks more accurate when training a data set where noise is added to the training examples, which forms the basis of differential privacy. [6]

Benefits:

1. Enhanced Privacy:
 - User Anonymity: Minimizes the chances that information such as a search performed or any activity detail can be linked to a unique user.
 - Data Security: Saves others from reverse engineering the model merely from the machine learning model outputs.
2. Regulatory Compliance:
 - Follows Privacy Laws: GDPR, CCPA regulative compliance, where user data is anonymized to meet specific legal compliance requirements.

- Eases Audits: This one is a clear benefit because it demonstrates the organization has put in place privacy measures, and tracking becomes easy during compliance audits.

Trade-offs:

1. Model Accuracy: The inclusion of noise can also have a negative impact in the sense that the accuracy achieved by the automatic systems shall be reduced.
2. Complexity: Unfortunately, the application of differential privacy is more sensitive to the control parameter and must be calibrated properly to ensure privacy and utility.

Therefore, to outweigh critics and abide by the rules, LinkedIn still needs to pursue constant enhancement and application of good privacy-preserving methods in their machine learning processes to grow the user's trust.

GOOGLE

1. Privacy and the need for Privacy-Preserving Machine Learning

Data Collected by Google.com:

1. Search History:
 - Record the searches you make through the Google Search Bar.
 - Aids refine a search and recommend other related content or advertisements for a user. [10]
2. Location Data:
 - Collects location data from Google Maps and GPS-enabled devices.
 - Used to provide location-based services like directions, local recommendations, and traffic updates. [10]
3. Email Content:
 - Analyzes Gmail data to filter spam and organize emails into categories (e.g., promotions, updates).
 - It helps improve the user experience by making email management easier. [10]
4. Device Information:
 - Some of the identified particulars include device types, the operating systems used, and browser history.
 - It is applied for the best performance and protection on various devices' systems. [10]
5. YouTube Activity:
 - Follow the search and watch history on YouTube.
 - It suggests videos and applies special filters to provide the best list of videos for you. [10]

Importance of privacy:

In Google, they used the use of ML in operations such as search engine optimization, advertising, filters against spam, and voice-controlled services. If stricter privacy measures are not put in place, then this data poses a high risk of data surveillance, leakage, and manipulation.

Potential Impact of Privacy Breaches:

1. Individual Users: Personal search history or email content and locations may expose the user to privacy infringement, harassment, and identity theft.
2. Society: That's why large-scale violations could lead to the manipulation of the population or the complete loss of confidence in technology companies.

Case study:

Again, in 2019, the company was fined €50 million under GDPR for not proactively offering the user enough transparency as to how his or her data is processed for personalized advertising. [11]

2. Achieving the Four Pillars of Privacy Preservation

Four Pillars of Privacy-Preserving AI on Google

1. Training Data Privacy: Google limits the data used for training machines, and even when it is collected, the data collected is anonymized and then aggregated.
2. Input Privacy: Aspects such as GMail end-user security with encryption throughout and incorporation of secure search arising out of private browsing.
3. Output Privacy: Applications of differential privacy include modifying the data of users in Gross, Amenities, etc., in the Google Map to provide general insights rather than of a specific individual.
4. Model Privacy: Opening research on federated learning for decentralized training of models and thus avoiding the sensitive data being centralized.

Recommendations for Improvement:

1. Federated Learning: The work should go on to ameliorate the decentralized ML models to reduce the data centralization risks.

2. Homomorphic Encryption: Extend the description of relevant algorithms about the advance in the required encrypted computations still higher to ensure absolute protection of the user inputs.
3. Secure Multi-Party Computation: Synchronize calculations between servers, allowing for data sharing without compromising data inputs.

3. Differential Privacy in Machine Learning

Google's use of differential privacy:

Differential privacy helps Google protect the user's information; they apply it while creating Google Maps or Android. It makes it possible for them to give forecasts and information such as traffic or location tendencies without violating personal details.

Benefits:

1. Enhanced Privacy:
 - Burdens individual user data by putting noise to mask information.
 - Also helps prevent such information from reaching a particular user.
2. User Trust:
 - Introduces users to the fact that the collected data is processed safely and appropriately.
 - Protects the user by creating more awareness of privacy services.

Trade-offs:

1. Accuracy: Including noise could further slightly alter the variability in search and recommendation procedures.
2. Complexity: It does prove to be a technically intensive model that demands some proportionate investment when rolled out at a large level.

Case Study:

Differential privacy is used by Google in generating COVID-19 mobility reports to analyze trends of location usage while maintaining the privacy of individuals' movement. This makes certain users' data remain private while others might be of some use to the one who is analyzing it. [12]

FACEBOOK

Privacy is important in social media companies such as Facebook.com because such platforms gather large amounts of user data for improving services via ML.

1. Privacy and the need for Privacy-Preserving Machine Learning

Data Collected by Facebook.com:

1. User Profiles:
 - Such as name, geographical location of user, and other preferences or even interests.
 - Facilitates Facebook to target you by recommending content or featured advertisements.
2. Social Connections:
 - Updates your friends list, groups, and your communications with other users.
 - People use this to find out what friends, groups, and events to suggest to you, depending on your network.
3. Content Sharing:
 - Saves the pictures, videos, and posts you upload in your profile.
 - Used to share your updates with your friends and to recommend content for you.
4. Messages:
 - Saves the text of private messages and communication.
 - Safeguards messages and enhances the chat features, such as who suggests.
5. Activity Logs:
 - Records activities that you perform, which include likes, comments, and shares on certain posts.
 - They assist in filtering your news feed and enhance plans and applications of targeted marketing.
6. Device Information:
 - Gathers information such as the kind of operating system the user is connected through, whether through a mobile device or a specific kind of Web browser.
 - It also aids in increasing performance and enhances an identical experience across devices.
7. Browsing Data:
 - Collects data on usage behavior from websites and/or applications that use the Facebook Like, comment, share, and other social plugins and custom audiences who interact with Facebook ads.

- Utilized for remarketing purposes and information about the audience's activity other than within Facebook.

Importance of Privacy:

Some of the applications of ML at Facebook are to filter feeds, recommend news feeds, advertise, and improve engagement. Keeping user data safe is critical to avoiding this kind of violation and losing control over personal information to fraudsters, competing firms, or other threat actors who wish to do the company and users harm.

Potential Impact of Privacy Breaches:

1. Individual Users: Persons' personal data, messages, or browsing history leakage may become an object of identity theft or a victim of a scandalous leak.
2. Society: This is ominous, especially when many users' data is implicated in large-scale breaches; this undermines confidence in SNSs. Besides, it disengages users and promotes disinformation.

Case Study:

The well-discussed case, the Facebook-Cambridge Analytics data scandal, was about collecting users' information without permission, mainly for political advertising. It proved a privacy leak and revealed a severe lack of protection, which shifted the focus on Facebook's data policy. [13]

2. Achieving the Four Pillars of Privacy Preservation

Four Pillars of Privacy-Preserving AI on Facebook

1. Training Data Privacy: Facebook has also created a process called Data Classification, which gives multiple data signals and machine learning to discover semantic types in data to improve privacy and security. [14]
2. Input Privacy: Looking at the need to safeguard the communication between two people through Messenger, end-to-end encryption should be adopted. [14]
3. Output Privacy: Investigating differential privacy approaches to prevent any revelation of personal information of users while in the generation of general information. [14]
4. Model Privacy: Investigation of methods for riding the need for training of the ML models while keeping the user data local to the user devices.

Recommendations for Improvement:

1. Federated Learning: Federated learning is thus extended to avoid aggregation of massive amounts of information, which enhances users' privacy. [14]
2. Homomorphic Encryption: Encrypt data and allow computations on this data using homomorphic encryption that does not let the data be compromised during computations. [14]
3. Secure Multi-Party Computation: Use secured multi-party computation to enable several parties to perform functions on their inputs without necessarily exposing those inputs. [14]

3. Differential Privacy in Machine Learning

Facebook has been researching differential privacy on how to improve the privacy of their machine learning algorithms. For example, they have studied how training examples could be made noisy more formally, a principal part of differential privacy. [15]

Benefits:

1. Enhanced Privacy:
 - Protects the user details so that people cannot track the personal information of any person.
 - Reduces spurious relationships, which might otherwise render sensitive information vulnerable to inference or misuse by the addition of noise.
2. Regulatory Compliance:
 - Controls user data to conform to privacy and protection regulations like the GDPR and CCPA.
 - Satisfies audit checks and acknowledges and complies with data protection regulations and guidelines everywhere in the world.

Trade-offs:

1. Model Accuracy: The inclusion of noise as a strategy inaccuracy affects machine learning models.
2. Complexity: While we consider differential privacy as a scalable privacy preservation solution, it should be noted that the level of privacy and utilities have to be properly adjusted to get good results.

Therefore, a lot more effort should be employed on the use of privacy-enhancing technologies in data processes with the help of machine learning in social media platforms, particularly Facebook, and this, coupled with the need to advance and employ better technologies, should enable the social media firm to ensure that users' privacy is upheld and adhered to due to the ever-changing privacy laws and regulations around the world.

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